

Course Code: BS1-EL-C1-T

Subject: ELECTRONICS –CT1

Title: ANALOG AND DIGITAL ELECTRONICS-I

Total No. of Teaching Hours: 54

No. of Hours per Week: 4

Internal Marks: 20

Exam Marks: 80

Credits: 3

Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT

Course Description:

This course provides a comprehensive introduction to fundamental electronic circuits and digital logic systems. It covers essential network theorems, semiconductor devices, rectifiers, voltage regulators, and the operation of bipolar junction transistors. Students will also explore number systems, Boolean algebra, and digital circuits including adders, subtractors, encoders, decoders, and multiplexers. Practical applications and circuit analysis techniques are emphasized to equip students with the skills needed for designing and analyzing electronic and digital systems.

Course Outcomes (Cos):

1. **Understand and Apply Network Theorems:** Students will understand and apply network theorems (KCL, KVL, Superposition, Thevenin's, Norton's, Maximum Power Transfer, and Reciprocity Theorems) to analyze and solve electrical circuits. (Bloom's Taxonomy Level: Understand, Apply)
2. **Analyze Semiconductor Devices and Rectifiers:** Students will analyze the operation and characteristics of PN junction and Zener diodes, and evaluate the performance of rectifiers and voltage regulators, including ripple factor and efficiency. (Bloom's Taxonomy Level: Analyze, Evaluate)
3. **Evaluate Transistor Circuits:** Students will evaluate the construction, operation, and biasing techniques of bipolar junction transistors, and apply this knowledge to analyze and design transistor-based amplifiers and feedback circuits. (Bloom's Taxonomy Level: Evaluate, Apply)
4. **Apply Boolean Algebra and Logic Design:** Students will apply Boolean algebra laws and analyze logic expressions using Karnaugh maps to design and simplify digital circuits. (Bloom's Taxonomy Level: Apply, Analyze)
5. **Design Digital Circuits:** Students will design and implement digital circuits including adders, subtractors, encoders, decoders, and multiplexers, and evaluate their performance and practical applications. (Bloom's Taxonomy Level: Design, Implement, Evaluate)
6. **Create Practical Electronic Systems:** Students will create and troubleshoot practical electronic and digital circuits, demonstrating their ability to apply theoretical knowledge to real-world problems and scenarios. (Bloom's Taxonomy Level: Create, Troubleshoot)

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1	1	1	2	3
CO2	3	1	1	1	2	2	3
CO3	3	2	1	1	2	1	3
CO4	3	1	2	1	2	1	3
CO5	3	2	2	2	2	1	3
CO6	3	2	2	1	3	1	3

UNIT-1

13 Hrs

Network Theorems: KCL & KVL, Superposition, Thevenin's, Norton's, Maximum Power Transfer, and Reciprocity Theorems. DC analysis of RC circuits, AC analysis of RLC series and parallel resonant circuits.

PN junction diode, Zener diode: Working, characteristics, and applications. Rectifiers: Half-wave and Full-wave rectifiers, expressions for output voltage, ripple factor, and efficiency (bridge rectifier), Shunt capacitor filter.

Voltage regulator. Line and Load regulation, Zener diode as voltage regulator - circuit diagram, load and line regulation, disadvantages. Fixed and Variable IC Voltage Regulators (78xx, 79xx, LM317), SMPS block diagram.

UNIT-2

15 Hrs

Bipolar Junction Transistor: Types, construction, working, and configurations, characteristics in CE mode, leakage currents, current gains (α , β), and their interrelations, DC load line, and Q point. Transistor as a switch. Transistor biasing: Thermal runaway, stability, and stability factor. Types of biasing.

Voltage Divider Bias. Amplifier: Classification, parameters, derivation for voltage and current gain of CE amplifier using model. Advantages of CC amplifier.

Multistage Amplifiers:

Two-stage RC Coupled Amplifier circuit, working, and its frequency response.

Large Signal amplifiers: Classification of large signal amplifier, ac load line, Class A, Class B and Class C power amplifiers (in brief)

UNIT-3

13 Hrs

Number System: Decimal, Binary, and Hexadecimal number systems, base conversions, representation of signed and unsigned numbers. Addition, subtraction, BCD code (8421), Gray code, error checking and correction codes, ASCII codes. Positive and negative logic, Boolean laws, Duality Theorem, De Morgan's Theorems, logic gates—AND, OR, NOT,

NAND, NOR, XOR & XNOR. Universal property of NOR and NAND gates. SOP and POS, Minterm, Maxterm, simplification of Boolean expressions, K-map for 3 and 4 variables.

UNIT-4

13 Hrs

Half Adder, Full Adder, Half Subtractor, Full Subtractor, 4-bit parallel binary adder. 3-bit magnitude comparator. Encoder: 4:2 encoder, decimal to BCD priority encoder (74147). Decoder: 2:4 decoder using AND gates, BCD to decimal decoder (7445), BCD to 7-Segment decoder (7446), Multiplexer 4:1 multiplexer, 1:4 De-multiplexer (logic diagram and truth table of each).

TEXT BOOK:

1. B Basavaraj, "Fundamentals of Electronics", Omkar Publications.
2. R.S. Sedha, "A Textbook of Applied Electronics," 7th edition, S. Chand and Company Ltd., 2011.
3. Thomas L. Floyd, "Digital Fundamentals," Pearson Education Asia, 1994.

Suggested References

1. Robert L. Boylestad, "Introductory Circuit Analysis," 5th edition, Universal Book, 2003.
2. A.P. Malvino, "Principles of Electronics," 7th edition, TMH, 2011.
3. Robert L. Boylestad and Nashelsky, "Electronic Devices and Circuit Theory," 11th edition, Pearson, 2013.
4. David A. Bell, "Electronic Devices and Circuits," 5th Edition, Oxford University Press, 2015.
5. A.P. Malvino, D.P. Leach, and Saha, "Digital Principles and Applications," TMH, 2011.
6. Anand Kumar, "Fundamentals of Digital Circuits," 2nd edition, PHI Learning Pvt. Ltd., 2009.

Course Code: BS1-EL-C1-P

Subject: ELECTRONICS –CP1

TITLE: ANALOG AND DIGITAL ELECTRONICS-I Lab

(Hardware implementation and Analysis of Circuit using Simulation Software)

Total No. of Teaching Hours: 52

No. of Hours per Week: 3

Internal Marks: 10

Exam Marks: 40

Credits: 2

Pedagogy: Problem Solving, Case studies, Seminar and using simulation tools

Course Description:

This course provides practical, hands-on experience with fundamental electronic components and circuit analysis techniques. Students will engage in experiments to understand and verify key electrical concepts, such as Thevenin's and Norton's theorems, rectification, and voltage regulation. The course covers the use of essential measuring instruments like multimeters, LCR meters, and oscilloscopes. Through practical experiments, students will become proficient in designing, testing, and troubleshooting electronic circuits, including rectifiers, voltage regulators, amplifiers, and digital logic circuits.

Course Outcomes (Cos):

1. **Component Identification and Measurement:** Students will be able to identify electronic components and measure their values using appropriate instruments, demonstrating proficiency in practical circuit analysis. (Bloom's Taxonomy Level: Understand, Apply)
2. **Circuit Analysis and Configuration:** Students will be able to analyse and measure various circuit configurations, including resistance, capacitors, and inductors in series, parallel, and series-parallel combinations. (Bloom's Taxonomy Level: Analyse, Apply)
3. **Effective Use of Measuring Instruments:** Students will demonstrate the ability to effectively use multimeters, LCR meters, and oscilloscopes for measuring voltage, current, resistance, capacitance, inductance, and frequency, and for verifying circuit performance. (Bloom's Taxonomy Level: Apply, Analyse)
4. **Verification of Network Theorems:** Students will verify and apply network theorems such as Thevenin's, Norton's, and Superposition in practical experiments, demonstrating their understanding of circuit analysis principles. (Bloom's Taxonomy Level: Evaluate, Apply)
5. **Design and Testing of Electronic Circuits:** Students will design, test, and analyse rectifiers, voltage regulators, and amplifiers, and assess their performance based on theoretical knowledge and practical results. (Bloom's Taxonomy Level: Design, Evaluate)
6. **Implementation of Digital Logic Circuits:** Students will design and implement digital logic circuits, including truth tables, adders, encoders, and decoders, and analyse their functionality and applications. (Bloom's Taxonomy Level: Create, Analyse)

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1	1	1	2	3
CO2	3	2	1	1	2	2	3
CO3	3	2	1	1	2	2	3
CO4	3	2	1	1	2	2	3
CO5	3	2	2	1	3	2	3
CO6	3	2	2	1	2	1	3

1. **Demonstration Experiments:**

Hands-on Experimental Skills and Familiarization with

- (a) Electronic components
 - (b) Resistance in series, parallel, and series-parallel combinations; Capacitors and Inductors in series and parallel
 - (c) Multimeter and LCR meter—checking of components
 - (d) Voltage sources in series, parallel, and series-parallel
 - (e) Voltage and current dividers
 - (f) Measurement of Amplitude, Frequency, & Phase difference using an Oscilloscope
2. Verification of Thevenin's Theorem.
 3. Verification of Maximum Power Transfer.
 4. Study of Series and Parallel Resonance circuits
 5. Study of the I-V Characteristics of a PN junction diode and Zener diode.
 6. Study of a half-wave rectifier without and with a shunt capacitor filter.
 7. Study of a full-wave bridge rectifier without and with a shunt capacitor filter.
 8. Study of the Zener diode as a Voltage Regulator.
 9. Study of Transistor characteristics in CE configuration.
 10. Study of single-stage CE amplifier.
 11. Study of two-stage RC-coupled CE amplifier.
 12. Verification of truth tables of OR, AND, NOT, NAND, NOR, XOR, and XNOR gates using NAND and NOR gates.
 13. Binary to Gray and Gray to Binary code conversion and parity checker using XOR IC 7486.
 14. Study of BCD to decimal decoder using IC7447.
 15. Study of the Encoders and priority encoders.
 16. Study of Multiplexer and Demultiplexer using ICs.
- Exposure to analog simulators LTspice, Circuit Logix, NI Multisim, CircuitMaker, Easy EDA, Every Circuit, PSpice, Docircuits, etc.

Course Code: BS2-EL-C2-T

Subject: ELECTRONICS –CT2

ANALOG AND DIGITAL ELECTRONICS-II

Total No. of Teaching Hours: 54

No. of Hours per Week: 4

Internal Marks: 20

Exam Marks: 80

Credits: 3

Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT

Course Description:

This course delves into advanced topics in semiconductor devices, including varactor, Schottky, tunnel diodes, and the applications of LED, LCD, and solar cells. Students will explore the working principles of various transistors such as JFET, MOSFET, IGBT, and thyristor-based devices like SCR, DIAC, and TRIAC. The course also covers operational amplifiers (Op-Amps), logic families, digital-to-analog and analog-to-digital converters, and sequential logic circuits. Emphasis is placed on practical applications and the design and analysis of registers, counters, and computer systems.

Course Outcomes (Cos):

1. **Understand and Apply Semiconductor Device Principles:** Students will be able to describe the working and applications of varactor, Schottky, and tunnel diodes, as well as LED, LCD, and solar cells. (Bloom's Taxonomy Level: Understand, Apply)
2. **Analyse and Compare Transistor Characteristics:** Students will be able to analyse the characteristics and working of JFETs, MOSFETs, and IGBTs, and compare them with BJTs in terms of their performance and applications. (Bloom's Taxonomy Level: Analyse, Compare)
3. **Design Circuits Using Thyristor Devices:** Students will design and test circuits involving SCR, DIAC, and TRIAC, demonstrating their ability to apply these devices in power control applications. (Bloom's Taxonomy Level: Design, Apply)
4. **Apply Operational Amplifiers in Various Configurations:** Students will design and analyse operational amplifier circuits, including inverting and non-inverting amplifiers, filters, and oscillators, and assess their performance in real-world applications. (Bloom's Taxonomy Level: Apply, Evaluate)
5. **Design and Analyse Digital Converters:** Students will design and implement digital-to-analog and analog-to-digital converters using binary weighted resistor and R-2R ladder networks, and evaluate their performance. (Bloom's Taxonomy Level: Design, Evaluate)
6. **Design and Analyse Sequential Logic Circuits:** Students will design and test sequential logic circuits such as flip-flops, counters, and shift registers, and demonstrate their application in memory devices and computer systems. (Bloom's Taxonomy Level: Analyse, Create)

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1	1	1	2	3
CO2	3	2	1	1	2	2	3
CO3	3	2	2	1	3	2	3
CO4	3	2	2	1	3	2	3
CO5	3	2	2	1	3	1	3
CO6	3	2	2	1	2	1	3

UNIT 1

12 Hrs

{**Self-Study:** Varactor diode, Schottky diode, Tunnel diode - LED, LCD, Solar Cell: Working and applications.}

JFET: Types, working, characteristics of n-channel JFET, parameters and their relationships, comparison of JFET and BJT.

MOSFET: Types, CMOS inverter, circuit and working.

IGBT: Construction and working, equivalent circuit and characteristics, intrinsic stand-off rating, Relaxation oscillator.

SCR: Working, characteristics, equivalent circuit, applications.

DIAC and TRIAC: Characteristics, equivalent circuit, working and applications for each.

UNIT 2

14 Hrs

Op-Amp: Differential Amplifier, Block diagram of Op-Amp, Characteristics of an Ideal and Practical Op-Amp, Concept of feedback—positive and negative—advantages and disadvantages. Open and closed loop configurations, Frequency Response, CMRR, Slew Rate and concept of Virtual Ground. Inverting amplifiers, Summing amplifier, Applications of Op-Amps: Inverter, Differentiator, Integrator, Logarithmic amplifier, Comparator.

Filters: First-order active Low pass, High pass, and Band pass Butterworth filters.

Oscillators: Barkhausen criterion for sustained oscillations, crystal oscillators, Phase Shift oscillator, Wien-Bridge oscillator using Op-amp.

IC 555 Timer: Astable and Monostable multivibrator circuits.

UNIT 3

14 Hrs

Logic Families: Pulse characteristics, Logic Families-classification of digital ICs. Characteristics of logic families, circuit description of TTL NAND gate with totem pole. TTL IC terminology. CMOS NAND, Comparison of TTL and CMOS families.

Digital to Analog Converter: DAC with binary weighted resistor and R-2R resistor ladder network.

Analog to Digital Converter: Successive approximation method—performance

characteristics.

Sequential Logic Circuits: Flip-Flops SR Latch, Level and Edge Triggered concepts, Clocked RS, D, JK, and T Flip-Flops. Preset and Clear operations. Race-around conditions in JK Flip-Flop. Master-Slave JK Flip-Flops. Applications of Flip-Flops.

UNIT 4

14 Hrs

Registers and Counters: Types of Shift Registers (up to 4-bits), their applications. Ring counter, Johnson counter applications. Asynchronous Counters: Logic diagram, Truth table and timing diagrams of 4-bit ripple counter, modulo-n counters, 4-bit Up-Down counter.

Synchronous Counters: 4-bit counter, Design of Mod 3, Mod 5, and decade Counters using K-maps.

Semiconductor memories: memory organization and operation, expanding memory size, Classification and characteristics of memories, RAM, ROM, EPROM, EEPROM, HDD, SSD.

Programmable logic devices: PAL, PLA, and PLDs, FPGA and CPLD

TEXT BOOK:

1. R.S. Sedha, "A Textbook of Applied Electronics," 7th edition, S. Chand and Company Ltd., 2011.
2. OP-Amps and Linear Integrated Circuit, R. A. Gayakwad, 4th edn., Prentice Hall., 2000
3. Thomas L. Floyd, "Digital Fundamentals," Pearson Education Asia, 1994.

Suggested References

1. Robert L. Boylestad, "Introductory Circuit Analysis," 5th edition, Universal Book, 2003.
2. A.P. Malvino, "Principles of Electronics," 7th edition, TMH, 2011.
3. Robert L. Boylestad and Nashelsky, "Electronic Devices and Circuit Theory," 11th edition, Pearson, 2013.
4. David A. Bell, "Electronic Devices and Circuits," 5th Edition, Oxford University Press, 2015.
5. A.P. Malvino, D.P. Leach, and Saha, "Digital Principles and Applications," TMH, 2011.
6. Anand Kumar, "Fundamentals of Digital Circuits," 2nd edition, PHI Learning Pvt. Ltd., 2009.

Course Code: BS1-EL-C1-P

Subject: ELECTRONICS –CP1

TITLE: ANALOG AND DIGITAL ELECTRONICS-II Lab
(Hardware implementation and Analysis of Circuit using Simulation Software)

Total No. of Teaching Hours: 52

No. of Hours per Week: 3

Internal Marks: 10

Exam Marks: 40

Credits: 2

Pedagogy: Problem Solving, Case studies, Seminar and using simulation tools

Course Description:

This laboratory course is designed to provide students with practical experience in the design, simulation, and analysis of analog and digital electronic circuits. Through hands-on experiments, students will explore the characteristics and applications of JFETs, UJT, SCRs, operational amplifiers, oscillators, and IC-555 timer-based circuits. The course also covers digital logic circuits, flip-flops, and digital-to-analog converters (DAC), providing an integrated understanding of both analog and digital electronics. Simulation software will complement the hardware experiments, enhancing circuit design and analysis skills.

Course Outcomes (Cos):

1. **Analyse JFET and UJT Characteristics:** Students will be able to study and determine the parameters of JFETs and UJT through practical experiments, demonstrating an understanding of their characteristics and applications. (Bloom's Taxonomy Level: Analyse, Apply)
2. **Design and Analyse Op-Amp Circuits:** Students will design and analyse various operational amplifier circuits such as inverting and non-inverting amplifiers, adders, differentiators, and integrators, and assess their frequency response and performance for different input waveforms. (Bloom's Taxonomy Level: Design, Evaluate)
3. **Design and Study Oscillator Circuits:** Students will design and study Wien Bridge, RC Phase Shift, and Crystal oscillators using operational amplifiers, and assess their performance based on theoretical expectations. (Bloom's Taxonomy Level: Design, Analyse)
4. **Implement and Test Timer Circuits:** Students will design and test ASTABLE and monostable multivibrators using IC-555 timers, and understand their use in practical timing applications. (Bloom's Taxonomy Level: Apply, Create)
5. **Implement Digital Logic Circuits:** Students will design and verify the operation of clocked RS, D, JK, and T flip-flops using NAND gates, demonstrating their understanding of sequential logic circuits. (Bloom's Taxonomy Level: Apply, Analyse)
6. **Design and Implement DACs:** Students will design a digital-to-analog converter using the binary weighted resistor method, and evaluate its performance in converting digital signals to analog output. (Bloom's Taxonomy Level: Design, Evaluate)

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	2	2	1	3
CO2	3	2	2	2	2	1	3
CO3	3	2	2	2	2	1	3
CO4	3	2	2	2	2	1	3
CO5	3	2	2	2	2	1	3
CO6	3	2	2	2	2	1	3

ELE-CP2: ANALOG AND DIGITAL ELECTRONICS-II Lab (Hardware and Circuit Simulation Software)

1. Study of JFET characteristics—determination of parameters.
2. Study of single-stage JFET amplifier.
3. UJT characteristics and relaxation oscillator.
4. SCR characteristics.
5. Design of inverting and non-inverting amplifiers using Op-amp & study of frequency response.
6. Op-amp inverting and non-inverting adder, subtractor.
7. Design and study of differentiator and integrator using Op-amp for different input waveforms.
8. Design and study of Wien Bridge oscillator using Op-amp.
9. Design and study of RC Phase Shift oscillator using Op-amp.
10. Design and study of first-order high-pass and low-pass filters using Op-amp.
11. Astable multivibrator using IC-555 timer.
12. Monostable multivibrator using IC-555 timer.
13. Digital to Analog Converter using binary weighted resistor method.
14. Study of Clocked RS, D, JK and T Flip-Flops using NAND gates
15. Study of decade counter using JK Flip-Flop.
16. Study of 4-bit Shift Register (SISO).

Using analog simulators (LTspice, CircuitLogix, NI Multisim, CircuitMaker, EasyEDA, EveryCircuit, PSpice, Docircuits, etc.), at least five experiments are to be performed.

Course Code: BS3-EL-CT3				Subject: ELECTRONICS –CT3																																																			
Title: EMBEDDED C AND DIGITAL DESIGN USING VERILOG																																																							
Total No. of Teaching Hours: 54					No. of Hours per Week: 4																																																		
Internal Marks: 20		Exam Marks: 80			Credits: 3																																																		
Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT																																																							
Course Description:																																																							
This course introduces the fundamentals of microcontrollers and embedded systems, focusing on the 8051 microcontroller, Embedded C programming, and interfacing with external devices. It covers key concepts of embedded system architecture, software development tools, and real-world applications using Arduino. The course also provides a comprehensive introduction to Hardware Description Languages (HDL) with Verilog, including different modeling styles (gate-level, dataflow, behavioral) and simulation techniques. Students will develop practical skills through programming, interfacing, and HDL design for embedded applications.																																																							
Course Outcomes (Cos):																																																							
CO1: Explain 8051 microcontroller architecture, embedded systems, and development tools.																																																							
CO2: Develop Embedded C programs for I/O operations, timers, counters, interrupts, and serial communication.																																																							
CO3: Interface sensors and actuators using Arduino and build basic embedded applications.																																																							
CO4: Describe Verilog HDL fundamentals and use simulation/synthesis tools.																																																							
CO5: Design and verify digital systems using gate-level, dataflow, and behavioural modelling in Verilog.																																																							
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)																																																							
<table><tr><td>COs</td><td>PO1</td><td>PO2</td><td>PO3</td><td>PO4</td><td>PO5</td><td>PO6</td><td>PO7</td></tr><tr><td>CO1</td><td>3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>2</td></tr><tr><td>CO2</td><td>3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>3</td></tr><tr><td>CO3</td><td>3</td><td>2</td><td>1</td><td>2</td><td>2</td><td>2</td><td>3</td></tr><tr><td>CO4</td><td>3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>3</td></tr><tr><td>CO5</td><td>3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>3</td></tr></table>								COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	CO1	3	2	1	1	2	1	2	CO2	3	2	1	1	2	1	3	CO3	3	2	1	2	2	2	3	CO4	3	2	1	1	2	1	3	CO5	3	2	1	1	2	1	3
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7																																																
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CO5	3	2	1	1	2	1	3																																																
UNIT-1							14 Hrs																																																
Introduction, Comparison between Microprocessors and Micro Controllers, RISC & CISC CPU Architectures, Harvard & Von-Neumann CPU architecture. Introduction to Embedded Systems, Difference between embedded and general-purpose computing, Examples of Embedded Systems, Design Parameters of Embedded Systems, Embedded Software Development Tools: Integrated Development Environment (IDE). Editor, Assemblers, Compilers, linker, loader, Instruction Set Simulator (ISS) Debugging Tools and Techniques, Emulators																																																							
The 8051 Architecture: Introduction to 8051 Microcontroller Hardware, Input/ Output Pins, Ports and Circuits- Port 0, Port1, Port2 and Port3.																																																							
External memory- Connecting external memory																																																							
Counters and Timers- Timer Counter Interrupts, Timing, Counting																																																							

UNIT-2	14 Hrs
Difference between C and Embedded C, Data types and time delays in 8051C, I/O programming, logic operations, data conversion programs, accessing code ROM space, data serialization. Programming 8051 Timers, Programming Counters, programming timers 0 and 1 in 8051C Interrupts Programming: 8051 Interrupts, Programming Timer Interrupts, External Hardware Interrupts. 8051 Serial Communication: Basics of Serial Communication, connections to RS 232C. Serial Communication Programming, Interrupt Programming. Arduino: Introduction to Arduino, types of Arduino, Arduino UNO, Arduino IDE, Arduino data types, function library, operators, control statement, loop statement, arrays, strings, interrupts. Arduino interfacing: Temperature and gas Sensor interfacing with Arduino, actuator interfacing (Preferably Step Motor) with an Arduino.	
UNIT-3	13 Hrs
Introduction to Verilog, Structure of HDL Module, Comparison of VHDL and Verilog Introduction to Simulation and Synthesis Tools, Test Benches. Language Elements- Keywords, Identifiers, Comments, format, Integers, reals and strings. Logic Values, Data Types-net types, undeclared nets, scalars and vector nets, Register type, Parameters. Delays, brief description - data flow style, behavioral style, structural style, mixed design style, simulating design. Expressions, Operands and Operators.	
UNIT-4	13 Hrs
Gate level Modeling: Introduction, built in Primitive Gates, multiple input gates, Tristate gates, pull gates, MOS switches, bidirectional switches, gate delay, array instances, and implicit nets. Data flow Modeling: Continuous assignment, net declaration assignments, delays, net delays. Behavioral Modeling: Procedural constructs, timing controls, block statement, procedural assignments, conditional statement, loop statement, procedural continuous assignment.	
TEXT BOOK: Suggested References 1. E. Balaguru swamy, "Computing fundamentals and C programming", 4th,Edition, TMH,2008. 2. Yashavant Kanetkar, "Let us C", 18th edition, BPB Publications, 2021. 3. Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D Mckinlay, "The 8051 Microcontroller and Embedded Systems", 2nd Edition, Pearson,2011 4. E. Balaguru swamy, "Programming in ANSI C", 2nd Edition.TMH, 2010. 5. Samir Palnitkar, "Verilog HDL: A guide to digital design and synthesis", Pearson, 2nd edition,2006. 6. J Bhasker, "A Verilog HDL Primer"" 3rd Edition, BS Publications,2008. 7. Nazesh M Botros, "HDL programming VHDL and Verilog". Dream tech press, 2009 reprint.	

Course Code: BS3-EL-CP3				Subject: ELECTRONICS –CP2																																																			
TITLE: EMBEDDED C & DIGITAL DESIGN USING VERILOG – LAB																																																							
Total No. of Teaching Hours: 52				No. of Hours per Week: 3																																																			
Internal Marks: 10			Exam Marks: 40			Credits: 2																																																	
Pedagogy: Problem Solving, Case studies, Seminar and using simulation tools																																																							
Course Description:																																																							
This laboratory courses focuses on hands-on experience with 8051 microcontroller programming using Embedded C on the Keil software platform and Arduino UNO interfacing for real-time applications. It also introduces digital circuit design using Verilog HDL and implementation on FPGA kits. Students gain practical skills in programming timers, serial communication, interrupts, interfacing sensors/actuators, and designing digital components such as adders, multiplexers, decoders, and counters using various modeling techniques.																																																							
Course Outcomes (Cos):																																																							
CO1: Develop Embedded C programs for 8051 microcontrollers using Keil software for timers, serial communication, and interrupts.																																																							
CO2: Interface sensors and actuators with Arduino UNO to build simple embedded applications.																																																							
CO3: Design and simulate digital logic circuits using Verilog HDL.																																																							
CO4: Implement combinational and sequential logic circuits on FPGA kits.																																																							
CO5: Apply programming and debugging skills to develop microcontroller-based and FPGA-based hardware systems.																																																							
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)																																																							
<table><tr><th>COs</th><th>PO1</th><th>PO2</th><th>PO3</th><th>PO4</th><th>PO5</th><th>PO6</th><th>PO7</th></tr><tr><td>CO1</td><td>3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>2</td></tr><tr><td>CO2</td><td>3</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>3</td></tr><tr><td>CO3</td><td>3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>2</td></tr><tr><td>CO4</td><td>3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>1</td><td>3</td></tr><tr><td>CO5</td><td>3</td><td>2</td><td>1</td><td>2</td><td>2</td><td>1</td><td>3</td></tr></table>								COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	CO1	3	2	1	1	2	1	2	CO2	3	2	2	2	2	2	3	CO3	3	2	1	1	2	1	2	CO4	3	2	1	1	2	1	3	CO5	3	2	1	2	2	1	3
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CO4	3	2	1	1	2	1	3																																																
CO5	3	2	1	2	2	1	3																																																
Section A: 8051 C Programming using KIEL software																																																							
1.	Toggling of ports.																																																						
2.	Experiments related to the timer in mode 1.																																																						
3.	Experiments related to timer in mode 2.																																																						
4.	Experiments related to serial data transfer with baud rate 4800.																																																						
5.	Experiments related to serial data transfer with baud rate 9600.																																																						
6.	Experiments related with interrupts.																																																						
7.	Interfacing Experiments using Arduino UNO(Any six experiments)																																																						

Section B: Digital Design Using Verilog with FPGA kit (Minimum SIX Programs is to be written and executed)	
1.	Realization of gates using Verilog code
2.	Realize Adder/Subtractor (Half) circuits using Verilog data flow description
3.	Realize Adder/Subtractor (Full) circuits using Verilog data flow description
4.	Realize the following code converters using Verilog behavioural description Gray to Binary and vice-versa Binary to excess 3 and vice-versa
5.	To realize 4-bit ALU using Verilog program.
6.	To realize using Verilog behavioural description: 8:1 multiplexer, 8:3 encoder.
7.	To realize using Verilog behavioural description: 1:8 DE multiplexer, 3:8 decoder
8.	To realize using Verilog behavioural description flip flops: (a) D-type (b) JK - type (c) T-type
9.	To realize counters: Up/down (Binary) using Verilog behavioural description.
10.	To realize counters: Up/down (BCD) using Verilog behavioural description.

Course Code: BS3-EL-DSE-3.1		Subject: ELECTRONICS –ELECTIVE-T3.1						
TITLE: RENEWABLE ENERGY AND ENERGY HARVESTING								
Total No. of Teaching Hours: 26					No. of Hours per Week:2			
Internal Marks: 10			Exam Marks: 40			Credits: 2		
Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT								
Course Description:								
This course introduces both conventional and non-conventional energy sources with a focus on their utilization patterns, environmental impacts, and future energy needs. It covers solar energy systems, photovoltaic applications, wind energy conversion, biomass energy processes, and emerging technologies like fuel cells and hydrogen energy. Students will gain knowledge of energy sector planning, renewable energy technologies, and their applications in sustainable power generation.								
Course Outcomes (Cos):								
CO1: Explain the characteristics, consumption trends, and environmental impact of conventional energy sources.								
CO2: Describe the principles, advantages, and limitations of various non-conventional energy sources.								
CO3: Analyse solar energy technologies, including solar cells, photovoltaic systems, and their applications.								
CO4: Evaluate wind and biomass energy systems, conversion processes, and their potential applications.								
CO5: Assess India’s energy scenario, renewable energy programs, and strategies for sustainable energy development.								
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)								
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
	CO1	3	2	1	1	2	2	2
	CO2	3	2	1	1	2	3	2
	CO3	3	2	1	1	2	3	2
	CO4	3	2	1	1	2	3	2
	CO5	3	2	2	2	2	3	3
UNIT 1							9 Hrs	
Conventional Energy sources: Coal, Oil, Natural Gas, Nuclear power and Hydro - their utilization pattern in the past, present and future projections of consumption pattern - Sector-wise energy consumption - environmental impact of fossil fuels - Energy scenario in India - Growth of energy sector and its planning in India. Non-Conventional Energy Sources: Solar Energy, Wind Energy, Energy from Biomass & Biogas, Ocean Thermal Energy Conversion, Tidal Energy, Geothermal Energy, Hydrogen Energy, Fuel Cell, Magneto Hydro-Dynamics Generator, Advantages& Limitations of Non- Conventional Energy Sources								
UNIT 2							9 Hrs	

Solar Energy: Solar cells for direct conversion of solar energy in to electric power, Solar cell parameter, Solar cell VI characteristics, Efficiency, single crystal silicon solar cells, polycrystalline silicon solar cells, Cadmium supplied solar cells, Application of solar energy- solar water heating - space heating and space cooling- solar photovoltaic cell - solar distillation - solar pumping - solar furnace- solar cooling - solar greenhouse. Photovoltaic applications: battery charger, domestic lighting, street lighting, water pumping etc. - solar PV power plant - Net metering concept, problems.

UNIT 3

9 Hrs

Wind and Bio mass energy: Nature of the wind - power in the wind - factors influencing wind - wind data and energy estimation -wind speed monitoring - wind resource assessment - Betz limit - site selection - wind energy conversion devices - classification, characteristics, applications.

Biomass resources and their classification - Biomass conversion processes - Thermo chemical conversion - direct combustion - biomass gasification - pyrolysis and liquefaction - biochemical conversion - anaerobic digestion - types of biogas Plants applications - alcohol production from biomass - bio diesel production - Urban waste to energy conversion - Biomass energy programme in India. Electricity observed in living systems, Problems.

References:

1. Non-Conventional Energy Sources, G.D. Rai, Khanna Publication.
2. Non-Conventional Energy Resources, B. H. Khan, The McGraw Hill Publishers. FLUID ECHANICS
3. Properties of Matter- Brijlal & Subramanyam, Eurasia Publishing House, 1991
4. Heat and Thermodynamics, D.S. Mathur, S.Chand & Co Ltd
5. Heat and Thermodynamics - Mark W.Zemansky and Richard H. Basic, McGraw Hill, New York, seventh edition.
6. Theory and experiment on Thermal Physics- P.K.Chakrabarti
7. Heat and Thermodynamics- Brijlal & Subramaniam, S.Chand
8. An Introduction to Mechanics, D. Kleppner and R. J. Kolenkow
9. Thermal Physics- Kittel, CBS Publishers, 1987.

Course Code: BS3-EL-DSE-3.2		Subject: ELECTRONICS –ELECTIVE-T3.2						
TITLE: POWER ELECTRONICS AND ELECTRIC VEHICLES								
Total No. of Teaching Hours: 26					No. of Hours per Week:2			
Internal Marks: 10			Exam Marks: 40			Credits: 2		
Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT								
Course Description:								
This course provides a foundation in power electronics with a special focus on its applications in electric vehicles (EVs). It covers power semiconductor devices, DC-DC converters, rectifiers, inverters, and PWM control techniques. Students will learn the fundamentals of EV architectures, components such as motors and batteries, and the role of power electronics in motor drives, regenerative braking, and EV charging systems.								
Course Outcomes (Cos):								
CO1: Explain the role of power electronics and semiconductor devices in modern applications, particularly in EV systems.								
CO2: Analyse the operation of DC-DC converters (buck, boost) and basic PWM control techniques.								
CO3: Describe AC-DC and DC-AC conversion topologies, inverter operation, and the use of SPWM for harmonic reduction.								
CO4: Understand EV fundamentals, architectures, motor types, batteries, and BMS functions.								
CO5: Evaluate the role of power electronics in EV applications such as motor drives, regenerative braking, charging systems, and emerging trends like V2G.								
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)								
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
	CO1	3	2	1	1	2	2	2
	CO2	3	2	1	1	2	2	2
	CO3	3	2	1	1	2	2	2
	CO4	3	2	1	1	2	3	2
	CO5	3	2	2	2	2	3	3
UNIT 1: Fundamentals of Power Electronics							9 Hrs	
Introduction to Power Electronics & Devices								
Overview: Role of power electronics in modem systems, specifically EVs.								
Power Semiconductor Devices: Diodes, MOSFETs & IGBTs V-I characteristics, switching behavior, key advantages in power applications, Brief mention of Wide Bandgap Devices (SiC, GaN) for EVs.								
Device Protection: Introduction to snubber circuits.								
DC-DC Converters								
Principles: Basic operation, role in voltage conversion.								
Step-Down (Buck) Converter: Circuit, operation (continuous mode), voltage conversion ratio.								
Step-Up (Boost) Converter: Circuit, operation (continuous mode), voltage conversion ratio.								
Control Basics: Introduction to Pulse Width Modulation (PWM).								
UNIT 2: AC Conversion and EV Fundamentals							9 Hrs	

AC-DC Converters (Rectifiers) Uncontrolled Rectifiers: Single-phase full-wave rectifier (bridge), concepts of average DC voltage, ripple. Controlled Rectifiers (Brief): Introduction to phase control using Thyristors (SCRs), basic concept. Power Factor Correction (PFC): Why it's needed, brief overview of active PFC. DC-AC Converters (Inverters) Introduction: Role in converting DC battery power to AC for motors. Single-Phase Inverters: Half-bridge and full-bridge topologies, square wave output. Three-Phase Inverters: Basic operation, 6-step waveform generation. Pulse Width Modulation (PWM) for Inverters: Sinusoidal PWM (SPWM) for harmonic reduction.	
UNIT 3 : Power Electronics in Electric Vehicles	9 Hrs
Electric Vehicle Fundamentals EV Classification: BEVs, HEVs, PHEVs, FCEVs- basic definitions and characteristics. EV Architectures: Simple block diagrams of series, parallel, and BEV powertrains. Key EV Components: Electric Motors: Introduction to PMSM and Induction Motors used in EVs. Batteries: Types (Li-ion focus), basic characteristics (voltage, capacity, C- rate), overview of Battery Management System (BMS) functions. Bidirectional DC-DC Converters: Importance in EV applications (e.g., HEVs, V2G). Power Electronics in EV Applications & Trends Motor Drives: Role of the inverter in controlling electric motors (basic concepts of speed control). Regenerative Braking: Principle and energy recovery. EV Charging Systems: On-board vs. Off-board charging, AC Charging (Level 1 & 2), DC Fast Charging (Level 3), Basic block diagram of a typical EV charger. Auxiliary Power: Brief mention of DC-DC converters for 12V/48V systems. Future Trends: Brief discussion on V2G and wide bandgap device adoption.	
References: 1. "Power Electronics: Converters, Applications, and Design" by Ned Mohan, Tore M. Undeland, William P. Robbins. Wiley. (Focus on first few chapters for devices and basic converters) 2. "Electric Vehicles: Principles, Design and Applications" by Iqbal Husain. CRC Press. (Focus on introductory chapters and powertrain)	

Course Code: BS4-EL-CT4			Subject: ELECTRONICS –CT4					
TITLE: ADVANCED COMMUNICATION SYSTEMS								
Total No. of Teaching Hours: 54			No. of Hours per Week: 4					
Internal Marks: 20		Exam Marks: 80		Credits: 3				
Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT								
Course Description:								
This course provides an in-depth understanding of advanced communication systems, including electromagnetic wave propagation, antennas, and transmission lines. It covers analog and digital modulation techniques, data transmission characteristics, and microwave devices like klystrons, magnetrons, and RADAR systems. Students will also gain insights into modern communication technologies such as satellite communication, optical fiber systems, and cellular networks, with emphasis on their components, principles, and practical applications.								
Course Outcomes (Cos):								
CO1: Explain the fundamentals of communication systems, electromagnetic wave propagation, and antenna parameters.								
CO2: Analyze various analog, pulse, and digital modulation techniques and their applications.								
CO3: Understand the principles and applications of microwave tubes, semiconductor microwave devices, and RADAR systems.								
CO4: Describe the working of satellite and optical fiber communication systems along with their components and characteristics.								
CO5: Illustrate the architecture and functioning of cellular communication systems, including frequency management, roaming, and encryption.								
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)								
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
	CO1	3	2	1	1	2	2	2
	CO2	3	2	1	1	2	2	2
	CO3	3	2	1	1	2	2	2
	CO4	3	2	1	1	2	2	2
	CO5	3	2	2	2	2	2	3
UNIT 1: Advanced Communication Systems							12 Hrs	
Introduction to communication – means and modes, Electromagnetic spectrum, band designations and usage, propagation of EM waves (Ground Wave, Sky-wave and Space-wave propagation), Introduction to communication system with general block diagram, Brief idea of frequency allocation for radio communication system in India (TRAI) and Analog Noise.								
Transmission Lines and Antennas:								
Transmission lines: Types of transmission lines and basic Principles (Fundamentals of Transmission lines, Characteristic Impedance, Losses in Transmission lines, Standing Waves),								
Antenna: Introduction, need for an antenna, radiation mechanism, Basic Types of Antennas, antenna parameters.								
Types of Antennas: Wire, Half-Wave Dipole, Half-Wave Folded Dipole, Rhombic, Loop, Helical								

Horn, Slot, Micro Strip, Lens and parabolic reflectors(Frequency Range, Construction & Working, Radiation Pattern, Advantages, Disadvantages and applications) Problems	
UNIT 2: Analog Pulse and Digital Modulation techniques	14 Hrs
Amplitude Modulation - representation, modulation index, instantaneous voltage, frequency spectrum, power relations. Limitations of AM. Frequency Modulation- representation, modulation index, frequency spectrum, bandwidth requirements, frequency deviation and carrier swing. Pulse Modulation- Types, PAM, PWM and PPM. Detailed discussion of PCM with its advantages, disadvantages and applications. Digital Communication: Block diagram of digital communication, Advantage and disadvantages of digital transmission systems. Characteristics of data transmission circuits-Shannon limit for information capacity. Bit Rate and Baud, bandwidth requirements, data transmission speed, noise, cross talk, echo suppressors, distortion and equalizer. Digital Modulation - Types (ASK, FSK, PSK, QPSK, 16QAM and 64 QAM), definition and waveforms and applications of each. MODEM- modes.	
UNIT 3 : Microwave Tubes, Semiconductor microwave devices and RADAR	14 Hrs
Introduction, Characteristic features of microwaves, Advantages and disadvantages of microwave, Applications of microwaves. Waveguides: Introduction, characteristics, advantages and types of waveguides. Comparison between transmission lines and waveguides. Microwave Tubes: Construction and working of Klystron amplifier, Reflex Klystron, Magnetron and Traveling Wave Tube (TWT). Semiconductor microwave devices: Gunn Effect and Gunn diode and tunnel diode. RADAR-Principles, Block diagram of pulsed RADAR system, RADAR range equation applications of RADAR.	
UNIT 4: Introduction to different types of Communication system:	13Hrs
Satellite Communications: Satellite orbits: Satellite elevation categories, satellite orbital patterns. Geosynchronous Satellites: orbital velocity, Clarke orbit, Advantages and disadvantages of geosynchronous satellites, satellite system link models: Satellite Multiple Accessing arrangements Optical Fiber Communication System: Block diagram of optical fiber communication system, optical fiber types, losses in optical fiber cables, light sources (LED and Laser diodes), light detectors (Photodiodes). Cellular Communication- Mobile communication, cell and cell splitting, frequency bands used in cellular communication, Absolute RF channel numbers (ARFCN), frequency reuse, roaming and hand off, authentication of the SIM card of the subscribers, IMEI number, data encryption, block diagram of cellular mobile communication network, Block diagram of cellular phone handset.	
TEXT BOOK: 1. Electronic Communication system, Kennedy & Davis, IV edition-TATA Mc-GRAW hill.	

2. Electronic Communication systems, Fundamentals through Advanced- Wayne Tomasi-Vth edition-Pearson Education

Suggested References

1. LAB experiments- Quarterly Journals, Kamaljeeth Instruments.
2. Principles of Communication Engineering-Umesh Sinha-Satya Prakashan, New Delhi
3. Advanced Communication System (communication system-II)-

Course Code: BS4-EL-CP4			Subject: ELECTRONICS –CP4																																																		
TITLE: Advanced Communication Systems - LAB																																																					
Total No. of Teaching Hours: 52			No. of Hours per Week: 3																																																		
Internal Marks: 10		Exam Marks: 40		Credits: 2																																																	
Pedagogy: Problem Solving, Case studies, Seminar and using simulation tools																																																					
Course Description:																																																					
This laboratory course provides hands-on experience with analog and digital communication systems. Students will perform experiments on modulation and demodulation techniques, pulse modulation methods (PAM, PWM, PPM), digital modulation schemes (ASK, BPSK, QPSK, DPSK), and time-division multiplexing (TDMA). The lab also covers optical fiber communication experiments, including measurement of losses, numerical aperture, and the characteristics of optical components. The practical sessions aim to strengthen the understanding of communication theory through real-world circuit implementation.																																																					
Course Outcomes (Cos):																																																					
CO1: Demonstrate the working of analog modulation and demodulation techniques (AM, FM).																																																					
CO2: Implement pulse modulation schemes (PAM, PWM, PPM) using discrete components and ICs.																																																					
CO3: Analyze digital modulation techniques (ASK, BPSK, QPSK, DPSK) and PCM systems.																																																					
CO4: Conduct experiments on optical fiber communication to study losses, numerical aperture, and optical components.																																																					
CO5: Apply time-division multiplexing and sampling theorem to real communication systems using hardware kits.																																																					
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)																																																					
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COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7																																														
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CO4	3	2	2	2	2	3	2																																														
CO5	3	2	2	2	2	3	3																																														
ELE-CP3: Advanced Communication Systems - LAB (Minimum Eight Experiments)																																																					
1.	Amplitude Modulator																																																				

2.	Amplitude demodulator
3.	Pre-Emphasis and De-Emphasis
4.	Voltage controlled oscillator using IC 555
5.	Frequency mixer
6.	Frequency Multiplier
7.	PAM using transistor
8.	PWM and PPM using IC-555
9.	ASK modulation and demodulation using OP-AMP or transistor
10.	Experiments on Optical Fibre (i) Characteristics of Optical fibre (ii) Study of losses in optical fibre. (iii) Study of numerical aperture of optical fibre. (iv) Study of characteristics of fibre optic LED's and photo detector.
11.	Sampling theorem using Communication kit
12.	BPSK, QPSK, DPSK
13.	TDMA
14.	PCM

Course Code: BS4-EL-DSE-4.1	Subject: ELECTRONICS –ELECTIVE-T4.1		
TITLE: COMPUTER NETWORKS			
Total No. of Teaching Hours: 26		No. of Hours per Week:2	
Internal Marks: 10	Exam Marks: 40		Credits: 2
Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT			
Course Description:			
This course introduces the principles and protocols of computer networks, focusing on the OSI and TCP/IP models. It covers key concepts of physical, data link, network, transport, and application layers, including error detection/correction, MAC protocols, IP addressing, and routing techniques. The course also explores transport layer protocols (TCP/UDP), congestion control, and application-level services like DNS, email, and WWW. Quality of Service (QoS) mechanisms and next-generation IP (IPv6) are also discussed.			
Course Outcomes (Cos):			
CO1: Explain protocol layering, OSI and TCP/IP models, and the fundamentals of physical and data link layers.			
CO2: Analyze error detection and correction methods, MAC techniques, and data link protocols like HDLC and PPP.			
CO3: Understand network layer protocols, IP addressing (IPv4/IPv6), and routing algorithms.			

CO4: Describe transport layer protocols (UDP/TCP), flow control, error control, and congestion control mechanisms.

CO5: Discuss application layer protocols (DNS, WWW, Email) and QoS techniques for improving network performance.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1	1	2	1	2
CO2	3	2	1	1	2	1	2
CO3	3	2	1	1	2	1	2
CO4	3	2	1	1	2	1	2
CO5	3	2	2	2	2	1	3

UNIT 1: Fundamentals and Internetworking

9 Hrs

Protocol Layering, The OSI Model, TCP/IP Protocol Suite, Introduction to Physical Layer: Transmission, Impairments, Data Rate Limits, Performance, Introduction to Data-Link-Layer: Link-Layer Addressing, Error Detection and Correction: Block Coding, Cyclic Codes, Checksum.

UNIT 2: Data Link and Network Layer

9 Hrs

Data Link Control: Data-Link Layer Protocols, HDLC, Point-To-Point (PPP), Media Access Control (MAC): CSMA, CSMA/CD, CSMA/CA, Reservation, Polling, Token Passing, FDMA, TOMA, CDMA, Network-Layer Services, Packet Switching, Network-Layer Performance, IPV4 Addresses, Network Layer Protocols: Internet Protocol (IP), ICMPv4, Mobile IP, Unicast Routing: Routing Algorithms, Unicast Routing Protocols, Next Generation IP: IPv6 Addressing

UNIT 3: Transport and Application Layer

9 Hrs

Introduction to Transport Layer: Introduction, Transport-Layer Protocols, Transport-Layer Protocols: User Datagram Protocol, Transmission Control Protocol: TCP Services, TCP Features, Segment, A TCP Connection, TCP Congestion Control, Flow Control, Error Control, Application Layer: WWW, E-MAIL, Domain Name System (DNS), Quality of Service: Flow Control To Improves QoS, Integrated Services

Computer Network Experiments

1. To study about components and specifications of Laptop and Desktop.
2. Installation and introduction of simulation tools packet tracer

3. Study of different types of network cables.
4. Familiarization with Transmission media and tools: Co-axial cable, UTP cable, Crimping tool, Connectors
5. Study of network IP address configuration (Classification of address, static and dynamic address)
6. Study of network devices (Switch, Router, Bridge)

References:

1. A.S Tanenbaum, Computer Networks, 4th Edition, PHI, 2003
2. Behrouz A. Foruzan, Data communication and Networking, 4th Edition, TMH, 2004
3. Larry L. Peterson and Bruce S. Davie, "Computer Networks A System Approach", 5th Edition, MKP, 2012.
4. James F. Kurose , Keith W. Ross, " Computer Networking, A Top-Down Approach", 5th Edition, Pearson, 2012.

Course Code: BS4-EL-DSE-4.2		Subject: ELECTRONICS –ELECTIVE-T4.2					
TITLE MEDITRONICS							
Total No. of Teaching Hours: 26					No. of Hours per Week:2		
Internal Marks: 10		Exam Marks: 40			Credits: 2		
Pedagogy: Classrooms lecture, Problem Solving, Case studies, Seminar and ICT							
Course Description:							
This course introduces the foundations of medical electronics, focusing on the role of electronic systems in healthcare, diagnostics, and therapy. It covers bioelectric signals (ECG, EEG, EMG), diagnostic and therapeutic devices, imaging modalities, and embedded systems in medical applications. Students will also learn about biomedical sensors, safety considerations, medical device standards, and emerging trends such as AI-driven diagnostics, smart wearables, and IoMT-based healthcare solutions.							
Course Outcomes (Cos):							
CO1: Explain the role of electronics in healthcare, bioelectric signals, and their acquisition using medical electrodes.							
CO2: Describe the design and working of diagnostic and therapeutic medical devices like ECG, EEG, EMG, pacemakers, and imaging subsystems.							
CO3: Analyze the operation of biomedical sensors, pulse oximeters, blood pressure monitors, and embedded medical control systems.							
CO4: Evaluate electrical safety, patient isolation, and medical device maintenance with respect to IEC and ISO standards.							
CO5: Discuss future trends in medical electronics including IoMT, AI-based diagnostic tools, and smart wearable devices.							
Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)							
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1	1	2	2	2
CO2	3	2	1	1	2	2	2
CO3	3	2	1	1	2	2	2
CO4	3	2	2	2	2	3	3
CO5	3	2	2	2	2	3	3
UNIT 1: Foundations of Medical Electronics:							9 Hrs
Role of electronics in healthcare and diagnostics, Classification of medical electronic devices: diagnostic, therapeutic, assistive, Bioelectric signals overview: ECG, EEG, EMG - characteristics and							

relevance, Electrode types and signal pickup: ECG, EEG, EMG probes, Signal conditioning essentials: amplifiers, active filters, isolation circuits, Noise sources and mitigation techniques in medical signal processing

UNIT 2

9 Hrs

Diagnostic and Therapeutic Electronics: Design and working of ECG, EEG, and EMG machines: block-level understanding Electronics in pulse oximeters and blood pressure monitors

Overview of imaging modalities: X-ray, CT, and ultrasound electronic subsystems Pacemakers and defibrillators: basic circuitry and operational principles, Infusion pumps and ventilators: embedded control and monitoring systems, Maintenance, calibration, and reliability in medical equipment

UNIT 3

9 Hrs

Embedded Systems and Safety in Medical Devices: Biomedical sensors: pressure, temperature, flow, oxygen saturation, Introduction to embedded systems in medical devices: Block diagram, Wireless health monitoring and the Internet of Medical Things (IoMT), Data acquisition systems for remote diagnostics, Electrical safety: leakage currents, patient isolation, fail-safes

Medical device standards: IEC 60601, ISO certifications, regulatory landscape Future trends: AI in diagnostic tools, smart wearables, and real-time analytics

References:


The National College
Bengaluru, Karnataka

Title: Communication II**Total No. of Teaching Hours: 52****No. of hours per week: 4****Internal Marks: 40****Exam Marks: 60****Credits: 4****Unit1:****10 hrs****Introduction to Spread Spectrum Communications:**

Frequency hopping spread spectrum (FHSS) systems, multiple fhss user systems and performance, applications of FHSS, direct sequence spread spectrum(DSSS), resilient features of DSSS, code division multiple-access (CDMA) of DSSS, multiuser detection (MUD).

Unit 2**14 hrs****Cellular Communication System:**

Introduction, Cellular Telephone, Frequency reuse, Interference, cell splitting, sectoring, segmentation and dualization, cellular system topology, roaming and handoffs, cellular telephone network components, cellular telephone call processing.

Authentication of the SIM card of the subscribers, IMEI number, concept of data encryption, architecture (block diagram) of cellular mobile communication network, simplified block diagram of cellular phone handset

Global system for mobile communications: GSM services, GSM system architecture, GSM radio subsystem, CDMA in Cellular Phone Networks.

Personnel satellite communications system: Advantages and disadvantages.

Unit 3**14 hrs****Introduction to wireless telecommunication systems:**

History and Evolution Different generations of wireless cellular networks 1G, 2G,3G and 4G and 5G technology and CV2X. Wireless network architecture and operation, wireless channel characteristics.

Network Connectivity: IEEE standards.

IEEE 802.15.4: Feature, variants, types, Frames, Beacon of Enabled Networks and Non-Beacon Enabled Networks.

Zigbee: Features, Components, Topologies, Types, Zigbee Network layer, Applications.

6LoWPAN protocol: Introduction, Features, Addressing, LOADng Routing, RPL Routing.

RFID: Introduction, Working Principle, Applications.

BLUETOOTH: Introduction, Working Principle, Applications.

Unit 4:**14 hrs****Data communications and networking:**

Introduction, Data communications network architecture, protocols and standards, Categories and topology of Networks,, standards organization for data communications, layered network architecture, open system interconnection, TCP/IP Suite, Serial and parallel data transmission, Data communications circuit arrangements, Data communications networks, alternate protocol suites

Data communications hardware, data communications circuits, Line control unit: UART and USRT, Serial interfaces: RS-232.

Public Switched Data Networks: Value added network, Packet switching network.

Integrated Services Digital Network: Principles, conceptual view, objective and architecture.

Asynchronous Transfer Mode: Header field, information field and network components.

Telephone and cable networks for data transmission, Telephone networks, Dial up modem, DSL, Cable TV for data transmission.

Text books:

1. Wayne Tomasi—"Electronic Communication Systems, Fundamentals through Advanced", 5 th edition.
2. "Digital Communications, Fundamentals and Applications", Bernard Sklar, 2/e, Pearson Education.
3. B. P. Lathi and ZhiDing, "Modern Digital and Analog Communication Systems", Oxford University Press,4thEdition, 2010

Reference Books:

1. D Roddy and J. Collen,"Electronics communications",4th edition, PHI,2008.
Bernard S k la 'Digital Communications: Fundamentals and Applications"
Pearson Education, 2 nd edition, 2009.
2. David T se,Pramod Viswanath 'Fundamentals of Wireless Communication',
Cambridge University Press,1stedition, 2005.
3. Wayne Tomasi "Advanced Electronic Communication Systems", -6 th edition,
Low priced edition-Pearson Education.
4. Kennedy & Davis, "Electronic Communication Systems", IVth edition-TATA
McGraw Hill.

ELE-CP5: Communication II Lab

Simulation Using MATLAB/SIMULINK or any other Simulation Software

1. Study of AM, FM, PM generation and Detection
2. Study of ASK generation and Detection
3. Study of FSK generation and Detection
4. Study of PSK generation and Detection
5. Active filter designing
6. BPSK, DPSK, QPSK modulator and demodulator
7. Study of PWM techniques
8. Pulse code modulation and demodulation system.
9. Computations of the Probability of bit error for coherent binary ASK, FSK and PSK for an AWGN Channel and compare them with their Performance curves.

Pedagogy: ICT lecture method, group discussion, seminar etc.

ELECTRONICS -CT6**Total No. of Teaching Hours: 52**

No. of hours per week: 4

Exam Marks: 60

Credits: 4

Embedded controller is a fundamental course for Electronics Students on the different types of controllers used in an embedded system. Embedded system play very important role in our day-to-day life.

- ✓ To know the importance of microcontrollers and its applications.
- ✓ Understand the basics of Embedded Systems hardware and software concepts.
- ✓ Acquire knowledge about 8051 and PIC Microcontrollers and its peripherals.
- ✓ Acquire knowledge about Arduino board and IDE.

At the end of the course the student should be able to:

- ✓ Identify and understand function of different blocks of 8051 microcontrollers.
- ✓ Develop program for I/O port operations, Timers, Serial port and Interrupts using C.
- ✓ Gain the knowledge to interface LCD, Keyboard, ADC, DAC, DC motor, etc.
- ✓ Do the projects using Arduino

Courses having focus on Employability/ Entrepreneurship/ Skill Development: This course is having focus on Skill Development.

[illegible]

Title: Embedded Controllers**Total No. of Teaching Hours: 52****No. of hours per week: 4****Internal Marks: 40****Exam Marks: 60****Credits: 4****Unit 1:****13 hrs****Introduction to Microcontrollers:**

Introduction, Comparison between Microprocessors and Micro controllers, RISC & CISC CPU Architectures, Harvard & Von-Neumann CPU architecture. Introduction to Embedded Systems, Examples of Embedded Systems, Design Parameters of Embedded Systems, Embedded Software Development Tools: Integrated development Environment (IDE). Editor, Assemblers, Compilers, linker, loader, Instruction Set Simulator (ISS) Debugging Tools and Techniques, Emulators

The 8051 Architecture: Introduction to 8051 Microcontroller Hardware, Input/Output Pins, Ports and Circuits- Port 0, Port1, Port2 and Port3.

External memory- Connecting external memory

Counters and Timers- Timer Counter Interrupts, Timing, Counting

Unit 2:**13 hrs****Addressing Modes and Instructions Sets:**

Introduction, addressing modes, External data Moves, Code Memory, Read Only Data Moves / Indexed Addressing mode, PUSH and POP Op-codes, Data exchanges, Example Programs; Byte level logical Operations, Bit level Logical Operations, Rotate and Swap Operations, Example Programs. Arithmetic Operations: Flags, Incrementing and Decrementing, Addition, Subtraction, Multiplication and Division, Decimal Arithmetic, Example Programs.

Jump and Call Instructions: The JUMP and CALL Program range, Jumps, calls and Subroutines, Interrupts and Returns, More Detail on Interrupts, Example Problems

Unit 3:**13 hrs****8051 programming in C**

Data types and time delays in 8051C, I/O programming, logic operations, data conversion programs, accessing code ROM space, data serialization.

Programming 8051 Timers, Counter Programming, programming timers 0 and 1 in 8051C

Interrupts Programming: 8051 Interrupts, Programming Timer Interrupts, External Hardware Interrupts.

Programming serial data transfer, Programming the Serial Communication Interrupts, Interrupt Priority in the 8051/52, Interrupt programming in C.

Interfacing with 8051: Interfacing 8051 to LCD, Keyboard, ADC, DAC, Stepper motor.

Unit 4:**13 hrs****PIC18 Microcontroller and Arduino:**

Overview of the PIC microcontroller family, Architecture and features of 18F458.

Arduino: Introduction to Arduino, types of Arduino, Arduino UNO, Arduino IDE, Arduino data types, function library, operators, control statement, loop statement, arrays, strings, interrupts.

Arduino interfacing: Temperature and gas Sensor interfacing with Arduino, actuator interfacing (Preferably Step Motor) with an Arduino.

TEXT BOOKS:

1. Kenneth J. Ayala ; “The 8051 Microcontroller Architecture, Programming & Applications” 2e, Penram International, 1996 / Thomson Learning 2005.
2. Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollin D.McKinlay; “The 8051 Microcontroller and Embedded Systems – using assembly and C ”- PHI, 2006 / Pearson, 2006

REFERENCE BOOKS:

1. Ajay V.Deshmukh; “Microcontrollers- Theory and Applications”,TMH,2005.
2. Predko ; “Programming and Customizing the 8051 Microcontroller” –, TMH
3. Tim Wilmshurst, “Designing Embedded Systems with PIC Microcontrollers:
4. Principles and applications”, First Edition, Elsevier, 2007.
5. Muhammad Ali Mazidi and Rolin D, Mckinlay, “PIC Microcontroller and Embedded Systems using assembly and C for PIC18,” 1stEdition, Pearson, 2008.
6. John Pitman, “Design with PIC Microcontrollers,” 1st Edition, Prentice Hall, 1997.

ELECTRONICS -CP6

Total No. of Teaching Hours: 52

No. of hours per week: 4

Exam Marks: 25

Credits: 2

It is aimed to provide the practical knowledge about the embedded controllers.

- ✓ To understand what is a microcontroller, microcomputer, embedded system..
- ✓ To understand different components of a micro-controller and their interactions.
- ✓ To understand programming environment used to develop embedded systems.
- ✓ To understand debugging techniques for an embedded system.

Upon successful completion of this course, students will be able to:

- ✓ Understand what is a microcontroller, microcomputer, embedded system.
- ✓ Understand different components of a micro-controller and their interactions.
- ✓ Become familiar with programming environment used to develop embedded systems
- ✓ Understand key concepts of embedded systems like IO, timers, interrupts, interaction with peripheral devices
- ✓ Learn debugging techniques for an embedded system

[illegible]

ELE-CP6: Embedded Controllers Lab
8051 PROGRAMMING using microcontroller 8051 kit and KEIL
(Any 10 Programs)

1. Program to add and subtract two 8-bit numbers.
2. Program to find 2's complements of an 16-bit number.
3. Program to find the sum of N 8-bit numbers.
4. Program to multiply two 8-bit numbers.
5. Program to multiply two 16-bit numbers.
6. Program to solve the linear equation $y = mx + c$.
7. Program to find the square of a number from look-up table.
8. Program to find largest of N numbers.
9. Program to find smallest of N numbers
10. Program to verify the truth tables of logic gates.
11. Program to find whether the given data is palindrome.
12. Program to arrange the numbers in ascending order.
13. Toggling of ports.
14. Experiments related with timer in mode1.
15. Experiments related with timer in mode2.
16. Experiments related with serial data transfer with baud rate 4800.
17. Experiments related with serial data transfer wit baud rate 9600.
18. Experiments related with interrupts.

Pedagogy: ICT lecture method, group discussion, seminar etc.

VI SEMESTER

Course code: BS6-EL-CT7

ELECTRONICS -CT7

Title: Electronic Instrumentation and Biomedical Instruments

Total No. of Teaching Hours: 52

No. of hours per week: 4

Internal Marks: 40

Exam Marks: 60

Credits: 4

Course Description:

This paper is divided into two parts: Electronic instruments and biomedical instrument. Biomedical instruments are the most important in day today's life.

Course Objective:

- ✓ Acquire the knowledge of various types of measurement systems and to know the importance of measuring instruments.
- ✓ To know different types of errors due to the measurement systems
- ✓ Gain the knowledge of working principle of sensors and actuators
- ✓ Understand the working principles of data acquisition systems

Course Outcomes (Cos):

At the end of the course the student should be able to:

- ✓ Able to calibrate the instruments to minimize measurement errors.
- ✓ Use different data acquisition systems to acquire real-time data
- ✓ Set up testing strategies to evaluate performance characteristics of different types of data acquisition system and develop professional skills in acquiring and applying the knowledge outside the classroom through design of a real-life instrumentation system.
- ✓ Apply/Investigate the diodes as rectifiers, Op-Amp, and logic gates.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

Courses having focus on Employability/ Entrepreneurship/ Skill Development: This course is having focus on Skill Development.

Course Outcomes (COs)	Program Outcomes (POs)								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	x								
CO2	x								
CO3	x								
CO4	x								
CO5	x								
CO6	x								
CO7	x								

Course code: BS6-EL-CT7

ELECTRONICS -CT7

Title: Electronic Instrumentation and Biomedical Instruments

Total No. of Teaching Hours: 52

No. of hours per week: 4

Internal Marks: 40

Exam Marks: 60

Credits: 4

BS6-EL-CT7: Electronic Instrumentation and Biomedical Instruments

Unit 1:

13 hrs

Introduction to instrumentation and transducers

Measurement Errors: Definitions of instrument, Accuracy, Precision, Resolution and Significant figures. Gross errors and systematic errors, Absolute and relative errors.

Whetstone's bridge, AC bridges, Capacitance Comparison Bridge, Inductance Comparison Bridge.

Transducers:

Transducers: Definition, Classification of transducers, Selecting a transducer, Construction, Principle of working and application of:

Displacement transducer: Capacitive transducer, Inductive transducer and LVDT

Temperature transducer: Thermocouples, Thermistors.

Photoelectric transducer: Photo conductive cells, Photovoltaic transducer.

Bolometer and RF power measurement using Bolometer.

Unit 2:

13 hrs

Sensors and actuators:

Sensors-Introduction to sensors, types of sensors, typical application of sensors, basic principles and operations of Thermal Sensors, Optical Sensors, Acoustic Sensors, MEMS, Nano-sensors, Ultrasonic Sensors, Thin Film Sensors, Liquid Level Sensors, Magnetic Sensors, Radiation Sensor

Actuators-Introduction to actuators, types of actuators, Logical and Continuous Actuators, Pneumatic actuator, Electro-Pneumatic actuator, cylinder, rotary actuators.

Unit3:

13 hrs

Fundamentals of biomedical instruments:

Sources of biomedical signals, Basic instrumentation system, General constraints in design of biomedical instrumentation systems.

Bioelectric Signals and Electrodes: Origin of bioelectric signals, Types of bioelectric signals, recording electrodes, Electrode-Tissue interface, Polarization, Skin contact impedance, Silver-silver chloride electrodes, Electrodes for ECG, EEG, EMG, Microelectrodes.

Electrocardiograph: Electrical activity of the heart, Genesis & characteristics of Electrocardiogram (ECG), Block diagram description of an Electrocardiograph, Electroencephalograph: Genesis of Electroencephalogram (EEG), Block diagram description of an Electroencephalograph.

Unit 4:

13 hrs

Medical Equipments:

Blood pressure measurement : Direct & Indirect method, Automatic blood pressure measuring apparatus using Korotkoff's method, Measurement of Respiration rate – Thermistor method, Impedance pneumography.

Cardiac Pacemakers and Defibrillators: Need for cardiac pacemaker, External pacemaker, Implantable pacemaker, Types of Implantable pacemakers, Programmable pacemaker, Rate-responsive pacemakers, and AC & DC defibrillators.

Specialized Medical Equipments: Electron microscope, X-ray tube, X-ray machine, Endoscopes, and thermography. Computerized axial tomography, Using ultrasound in medicine, Ultrasound scanning, Magnetic resonance imaging, PET and SPECT Imaging

Text books:

1. “Modern Electronic Instrumentation and Measurement Techniques”, A.D. Helfric and W.D. Cooper, PHI of India ltd.,
2. Bio-Medical instrumentation- Rangan Mani Sharma
3. Bio-Medical Instrumentation- Dr. M Arumugam

BS6-EL-CP7- Instrumentation and IoT Lab

1. Displacement measurement using LVDT.
2. Temperature measurement using RTD, Thermistor and Thermocouple, and finding their sensitivity.
3. Temperature measurement using AD590/LM34.
4. Characteristics of LDR, Photodiode & Phototransistor by variable illumination & variable distance.
5. Measurement of unknown resistance by Wheatstone bridge & finding the sensitivity of the bridge.
6. Measurement of self-inductance using Maxwell's bridge.
7. Measurement of unknown capacitance using Schering's bridge.
8. Characteristics of pressure transducer.
9. Characteristics of potentiometric transducer.
10. Conversion of basic ammeter into multimeter and calibration of voltage, resistance and current.
11. Interfacing Experiments using Arduino UNO

Pedagogy: ICT lecture method, group discussion, seminar etc.

Course code: BS6-EL-CT8

ELECTRONICS -CT8

Title: Internet of Things and Robotics

Total No. of Teaching Hours: 52

No. of hours per week: 4

Internal Marks: 40

Exam Marks: 60

Credits: 4

Course Description:

This paper is divided into two parts: Electronic instruments and biomedical instrument. Biomedical instruments are the most important in day today's life.

Course Objective:

- ✓ Acquire the knowledge of various types of measurement systems and to know the importance of measuring instruments.
- ✓ To know different types of errors due to the measurement systems
- ✓ Gain the knowledge of working principle of sensors and actuators
- ✓ Understand the working principles of data acquisition systems

Course Outcomes (Cos):

At the end of the course the student should be able to:

- ✓ Able to calibrate the instruments to minimize measurement errors.
- ✓ Use different data acquisition systems to acquire real-time data
- ✓ Set up testing strategies to evaluate performance characteristics of different types of data acquisition system and develop professional skills in acquiring and applying the knowledge outside the classroom through design of a real-life instrumentation system.
- ✓ Apply/Investigate the diodes as rectifiers, Op-Amp, and logic gates.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

Courses having focus on Employability/ Entrepreneurship/ Skill Development: This course is having focus on Skill Development.

Course Outcomes (COs)	Program Outcomes (POs)								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	x								
CO2	x								
CO3	x								
CO4	x								
CO5	x								
CO6	x								
CO7	x								

Course code: BS6-EL-CT8

ELECTRONICS -CT8

Title: Internet of Things and Robotics

Total No. of Teaching Hours: 52

No. of hours per week: 4

Internal Marks: 40

Exam Marks: 60

Credits: 4

BS6-EL-CT8: Internet of Things and Robotics

UNIT 1:

13 hrs

IoT and Wireless Sensor networks

Introduction to IoT: Introduction, Origin of terminology, Characteristics, Evolution of connected devices, Modern Day IoT applications, Baseline Technologies (Machine to machine(M2M) communications, cyber physical systems, Web of Things (WoT))

IoT protocols: MQTT, UDP, HTTP, COAP, XMPP and gateway protocols (All the protocols in brief)

Industrial IoT: Industrial IoT Case Study: Agriculture, Healthcare, Activity Monitoring

Wireless Sensor Networks: Target tracking, Wireless Multimedia Sensor Networks (WMSNs), Stationary Wireless Sensor Networks (SWSN), Mobile Wireless Sensor Networks (MWSN), UAV Networks.

UNIT 2:

13 hrs

Control System

Functional elements of control system, Continuous and discrete state control systems, Open and closed loop systems.

Laplace transform in control systems:

Mathematical models of systems, transfer function, Block diagram of the control system and rules for its reduction, Signal flow graphs, Mason's gain formula. Analysis of multiple input multiple output systems.

Time-Domain Analysis: Test input signal for transient analysis, Time domain performance criterion, Transient response of first order, second order, and higher order systems, sinusoidal steady state analysis. Error analysis: Static and dynamic error coefficients, Error criterion, Introduction to system optimization.

Unit 3:

13 hrs

Introduction to ROBOTICS

Basic concepts, need, law, history, anatomy, specification.

Robot Configurations-Cartesian, cylinder, polar, and articulate.

Introduction to robot control, Control schemes used for robotic manipulator-joint motion control, Resolved motion control, and Adaptive control schemes.

Robot elements: end effectors classification, types of mechanical actuation, gripper design, robot drive system, types, position and velocity feedback devices, Robot wrist mechanism, precision, and accuracy of the robot.

UNIT 4:

13 hrs

Robot kinematics:

Basics of direct and inverse kinematics, robot trajectories, 2D and 3D. transformation-scaling, rotation, translation, homogeneous transformation.

Robot application: industrial applications of robots, medical, household, entertainment, space, underwater, defence, and disaster management. Micro and robots

Text Books:

1. Alessandro Bassi, Martin Bauer, Martin Fiedler, Thorsten Kramp, Rob van Kranenburg, Sebastian Lange, Stefan Meissner, “Enabling things to talk – Designing IoT solutions with the IoT Architecture Reference Model”, Springer Open, 2016.
2. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stamatis Karnouskos, Stefan Avesand, David Boyle, “From Machine to Machine to Internet of Things”, Elsevier Publications, 2014.
3. Fundamentals of Robotics by D K Pratihar.
4. Robotics Simplified: An Illustrative Guide to Learn Fundamentals of Robotics, by Dr. Jisu Elsa Jacob, Manjunath N.
5. Introduction to Robotics | Fourth Edition by John Craig.
6. Arduino Robotics by John-David Warren (Author), Josh Adamsduino.

Course code: BS6-EL-CP8

ELECTRONICS -CP8

Title: Project Lab

Total No. of Teaching Hours: 52

No. of hours per week: 4

Internal Marks: 25

Exam Marks: 25

Credits: 2

Course Description:

This course is about the project

Course Objective:

- ✓ To make the students practically sound.

Course Outcomes (Cos):

At the end of the course the student should be able to:

- ✓ Do the project
- ✓ Understand the working of each components used in the project.
- ✓ Get keen knowledge about the subject.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (Pos)

Course Outcomes (COs)	Program Outcomes (POs)								
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1									
CO2	x								
CO3	x								
CO4									
CO5	x								
CO6									
CO7									
CO8									

BS6-EL-CP8- Project Lab

Guidelines for evaluation of Projects:

Project must be completed in the department in the supervision of faculty in charge. It must incorporate the fundamental concepts of Electronics and use the latest tools available to design the need of the problem. Scope for improving the model must be specified in conclusion along with detailed report on the project.

Department must conduct periodical seminars during the allotted lab hours to check the progress of the project. Internal assessment marks of 25 can be awarded based on the performance of the students in the regular lab hours, Seminar presentations, type of the project and application of the project.

Pedagogy: ICT lecture method, group discussion, seminar etc.